

End-to-End Data Science Project Report

Titanic Survival Prediction Using Machine Learning

Internship Project

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Technology Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

1. Introduction

Data Science is the process of extracting meaningful information from data to support better decision-making. A complete data science project involves collecting data, cleaning it, analyzing patterns, building predictive models, evaluating their performance, and communicating the results effectively.

In this project, the Titanic dataset is used to build a machine learning model that predicts whether a passenger survived or not. The project follows the complete end-to-end data science workflow, starting from raw data and ending with model evaluation and business insights.

2. Problem Statement

The sinking of the Titanic remains one of the most well-known maritime disasters in history. The objective of this project is to analyze passenger information and predict whether a passenger survived based on factors such as age, gender, passenger class, fare, and family size.

This is a binary classification problem where the target variable is:

- **1 = Survived**
 - **0 = Did Not Survive**
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3. Objectives

The objectives of this project are:

- Load and understand the Titanic dataset.
 - Clean and preprocess the data.
 - Handle missing values.
 - Perform Exploratory Data Analysis (EDA).
 - Create useful features for model training.
 - Train a Machine Learning classification model.
 - Evaluate model performance.
 - Generate meaningful insights and recommendations.
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4. Dataset Description

The Titanic dataset contains information about passengers travelling on the Titanic.

The dataset includes information such as:

- Passenger Class
- Passenger Name
- Gender
- Age
- Number of Siblings/Spouses
- Number of Parents/Children
- Ticket Number
- Fare
- Cabin
- Embarked Port
- Survival Status

Several columns such as Passenger ID, Name, Ticket, and Cabin were removed because they were not useful for prediction.

5. Data Cleaning and Preprocessing

Before building the machine learning model, the dataset required preprocessing to improve data quality.

The following preprocessing steps were performed:

- Removed unnecessary columns.
- Filled missing values in the Age column using the median value.
- Filled missing values in the Embarked column using the most frequent value (mode).

- Converted categorical variables into numerical values using Label Encoding.
- Created a new feature called **FamilySize** by combining siblings, spouses, parents, and children.

These preprocessing steps ensured that the dataset was complete and suitable for machine learning.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns within the dataset.

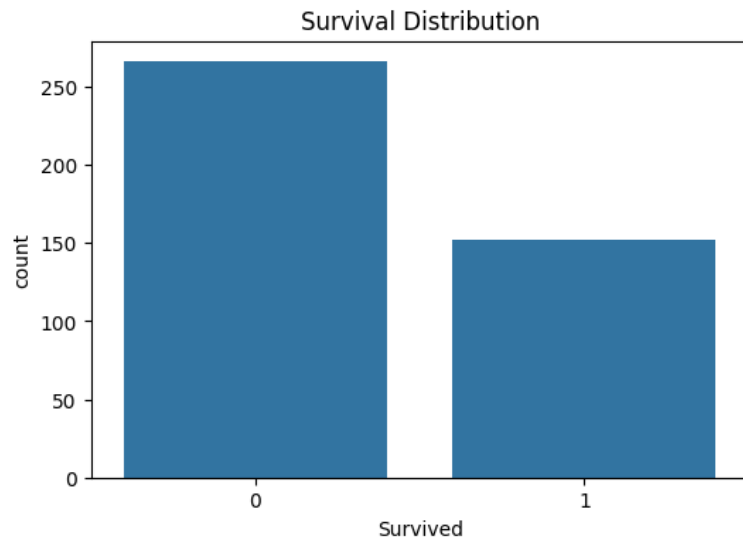
The following visualizations were created:

Missing Values Heatmap



This visualization highlighted missing values before data cleaning and confirmed which columns required preprocessing.

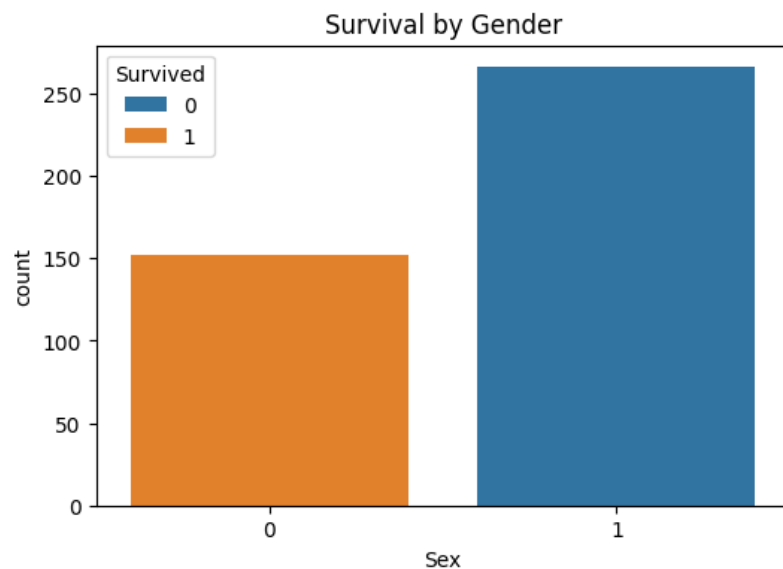
Survival Distribution



This chart showed the number of passengers who survived compared to those who did not.

It helped understand the class balance in the dataset.

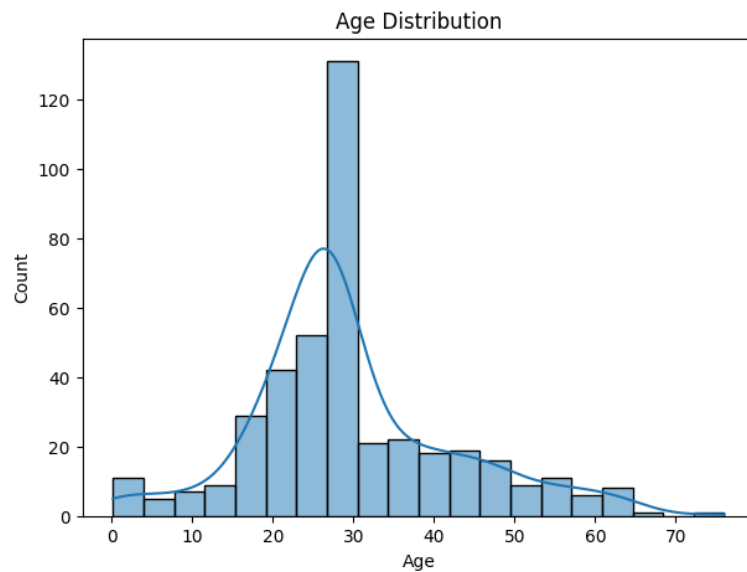
Survival by Gender



This visualization compared survival rates between male and female passengers.

It showed that female passengers had a significantly higher survival rate.

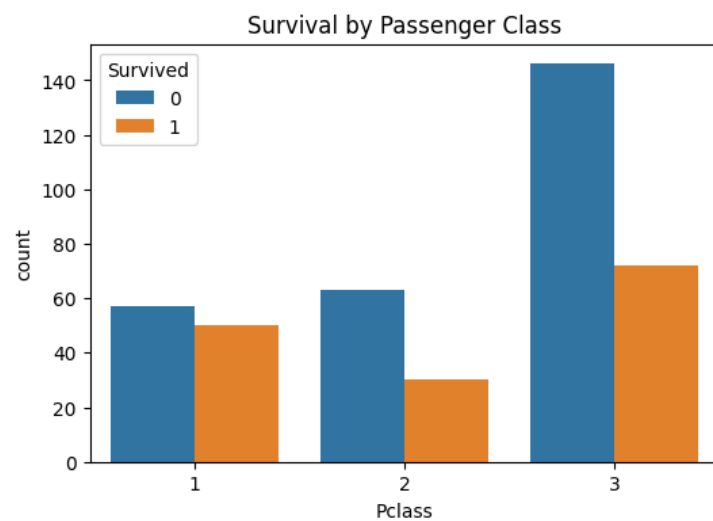
Age Distribution



The histogram illustrated the distribution of passenger ages.

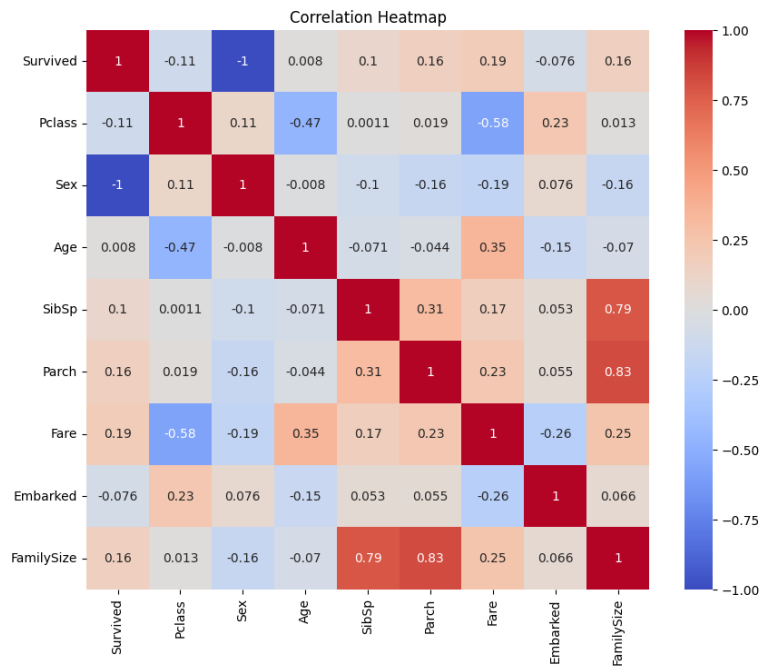
Most passengers were between 20 and 40 years old.

Survival by Passenger Class



Passengers travelling in First Class showed higher survival rates compared to passengers travelling in Third Class.

Correlation Heatmap



The heatmap displayed relationships between numerical variables.

It helped identify features that were positively or negatively associated with survival.

7. Feature Engineering

Feature engineering was performed to improve the predictive ability of the model.

The following transformations were applied:

- Label Encoding for categorical features.
- Creation of a new feature named **FamilySize**.

FamilySize was calculated as:

$$\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$$

This feature represents the total number of family members travelling together.

8. Machine Learning Model

A Random Forest Classifier was selected for this project.

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their predictions.

The dataset was divided into:

- **80% Training Data**
- **20% Testing Data**

The model was trained using the training dataset and evaluated using the testing dataset.

9. Model Evaluation

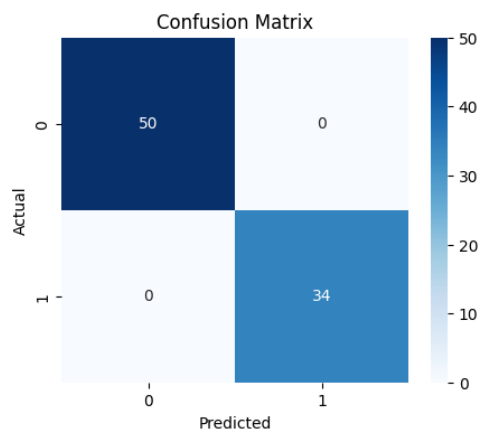
The trained model was evaluated using standard classification metrics.

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

These metrics provide a comprehensive understanding of model performance.

The confusion matrix was also plotted to visualize correct and incorrect predictions.



10. Results

The Random Forest model achieved good classification performance on the Titanic dataset.

The model successfully identified important relationships between passenger characteristics and survival.

Feature Importance analysis revealed that variables such as:

- Gender
- Passenger Class
- Fare
- Age
- Family Size

had the greatest influence on predicting survival.

11. Key Insights

The analysis produced several important findings:

- Female passengers had a much higher survival rate than male passengers.
- First-class passengers were more likely to survive than third-class passengers.
- Younger passengers generally had better survival chances.
- Family size influenced survival probability.
- Passenger class and gender were among the strongest predictors of survival.

These insights demonstrate how demographic and socioeconomic factors influenced survival during the Titanic disaster.

12. Business / Real-World Interpretation

Although the Titanic dataset represents a historical event, the techniques used in this project have practical applications in many industries.

Examples include:

- Customer churn prediction
- Loan approval systems
- Fraud detection
- Medical diagnosis
- Insurance risk assessment
- Employee attrition prediction

The project demonstrates the complete lifecycle of building predictive models for real-world decision-making.

13. Challenges Faced

Several challenges were encountered during the project:

- Missing values in the dataset.
- Presence of categorical variables.
- Feature selection.
- Creating meaningful features.
- Evaluating model performance effectively.

Each challenge was addressed using appropriate preprocessing and machine learning techniques.

14. Conclusion

This project successfully demonstrated the complete end-to-end data science workflow using the Titanic dataset.

The project included data loading, preprocessing, exploratory data analysis, feature engineering, model building, evaluation, and insight generation.

The Random Forest Classifier effectively predicted passenger survival using demographic and travel-related information.

The project strengthened practical knowledge of machine learning, data preprocessing, visualization, and model evaluation while demonstrating how data science can be used to solve real-world classification problems.

15. Future Improvements

The following improvements can be made in future versions of the project:

- Compare additional machine learning algorithms such as Logistic Regression, XGBoost, and Support Vector Machines.
- Perform hyperparameter tuning using Grid Search.
- Deploy the model using Flask and Docker.
- Build an interactive web application using Streamlit.
- Deploy the application to cloud platforms such as AWS or Google Cloud.

Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Project Workflow

